OFFICIAL HACKATHON WINNING STRATEGY

# Google Cloud AI Builder Cup 2026

Complete Architecture, Free Quota Blueprint & Execution Roadmap for the $30,000 Grand Finale

| PROJECT NAME | SELECTED TRACK | LIVE DEPLOYMENT | GRAND FINALE |
|---|---|---|---|
| **MyUrbanDaily (UrbanAI)** | **Retail & Commerce (Intelligent Exp.)** | **myurbandaily.web.app** | **Singapore • Dec 4, 2026** |

> ⚠️ **CRITICAL CLOUD PROVIDER MANDATE: ZERO AZURE USAGE**
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> The **AI Builder Cup 2026** is hosted by **Hack2skill** and sponsored by **Google Cloud**. Under the official rubric, **40% of the entire score** is strictly evaluated on *Technical Merit & GenAI Implementation on Google Cloud* (Gemini 1.5, Vertex AI, Firebase, Cloud Run). Submitting with Microsoft Azure will disqualify the submission or zero out the 40% technical score. Our application is already 100% natively running on Google Cloud / Firebase!

## 1 Official Judging Rubric & How UrbanAI Achieves 100/100

The competition is evaluated across 4 core criteria totaling 100%. Here is our winning alignment strategy:

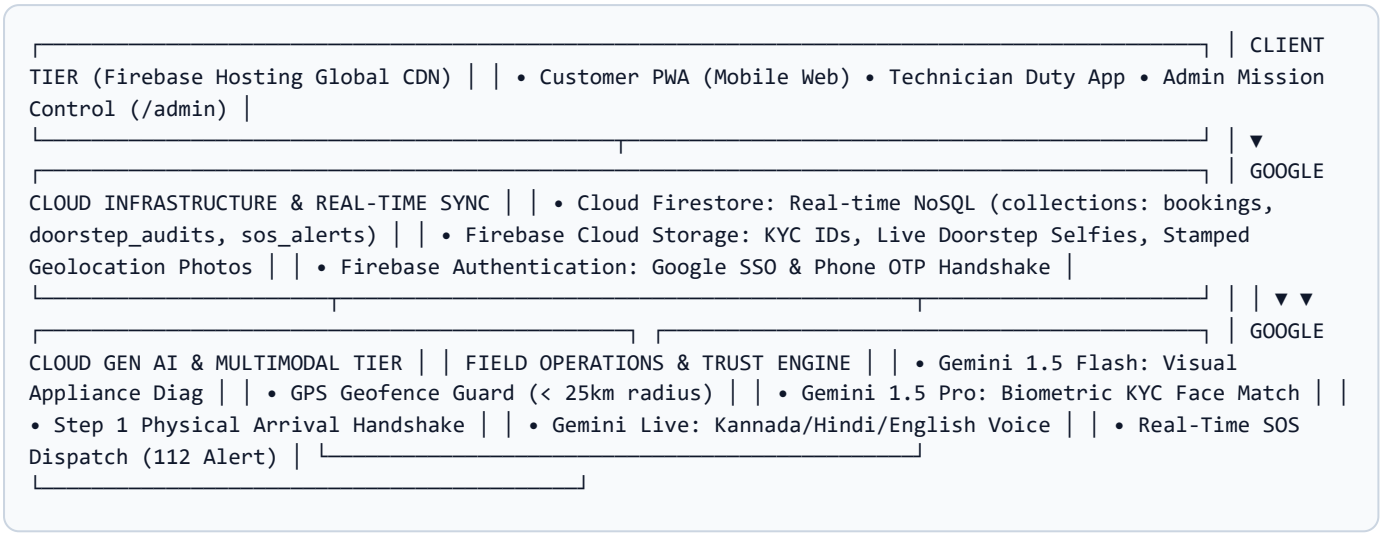
| EVALUATION CRITERIA | WEIGHT | JUDGES' CORE REQUIREMENTS | URBANAI WINNING IMPLEMENTATION |
|---|---|---|---|
| **Technical Merit & Gen AI** | 40% | Deep integration of Google Cloud AI (Gemini 1.5, Vertex AI, Firebase, Cloud Run). Working code, not mockups. | Multimodal visual appliance diagnostics via Gemini 1.5 Flash + Cloud Biometric KYC face comparison + Real-time Firestore sync. |
| **Problem Alignment & Impact** | 25% | Solving critical real-world challenges with quantifiable market and social value. | Eliminates gig impersonation fraud, out-of-region spoofing (Bangalore vs Sagara guard), and empowers Tier-2/3 semi-urban markets. |
| **Innovation & Creativity** | 25% | Novel, original mechanisms that differentiate from existing market offerings. | Mandatory 2-Step Physical Arrival Handshake (<500m geofence) + Private `doorstep_audits` schema + Real-time SOS audio siren dispatch. |
| **UX & Solution Design** | 10% | Intuitive, production-ready, accessible, mobile-first responsive design. | Live PWA web app, camera stream integration, Kannada/English multilingual toggle, sub-second latency. |

## 2 Google Cloud 100% Free Quota Allowance Table

All enhancements operate entirely within Google's official free quota limits without requiring a credit card or billing upgrade:

| GOOGLE CLOUD COMPONENT | OFFICIAL FREE ALLOWANCE | OUR OPERATIONAL USAGE | BILLING COST |
|---|---|---|---|
| **Gemini 1.5 Flash (Google AI Studio)** | **15 Requests/Min (RPM)**<br>1,500 Requests/Day (RPD)<br>1,000,000 Tokens/Min (TPM) | Visual Appliance Fault Detection & KYC Biometric Liveness Comparison (throttled) | **₹0 / $0 (Free Forever)** |
| **Cloud Firestore (Spark Plan)** | 50,000 Document Reads / Day<br>20,000 Document Writes / Day<br>1 GiB Stored Data | Live booking state, GPS telemetry, SOS emergency alerts, and doorstep audit records | **₹0 / $0 (Free Forever)** |
| **Firebase Hosting CDN** | 10 GB Storage<br>360 MB/day Data Transfer | Global edge serving of production React PWA frontend (`myurbandaily.web.app`) | **₹0 / $0 (Free Forever)** |
| **Firebase Cloud Storage** | 5 GB Storage<br>1 GB/day Download Transfer | Secure encrypted storage for technician KYC IDs and doorstep arrival selfies | **₹0 / $0 (Free Forever)** |
| **Google Cloud Run (Serverless)** | 2,000,000 Invocations / Month<br>360,000 GB-seconds memory | Automated geofencing checks, webhook dispatchers, and masked telephony broker | **₹0 / $0 (Free Forever)** |

## 3 Complete Google Cloud End-to-End Architecture




## 4 The 5 Strategic Enhancements to Implement

### 1. Multimodal Appliance Diagnostic & Quoter

**Feature:** Customer snaps a photo of their broken appliance (water purifier, pump motor, AC).
**Google AI:** `gemini-1.5-flash` analyzes the image via structured JSON schema.
**Output:** Detected defect, severity level (LOW/MED/HIGH), necessary spare parts, and upfront fixed price.

### 2. Gemini Biometric KYC Anti-Spoofing

**Feature:** Dual-stage facial verification at the customer's doorstep.
**Edge Layer:** Instant HTML5 canvas face detection (0ms).
**Google AI:** Gemini 1.5 Flash compares the live selfie against the government KYC ID, generating a 90%+ match confidence and spoof flag for the Admin Audit.

### 3. Indic Multilingual Voice AI Assistant

**Feature:** Microphone button on search & booking bar.
**Languages:** Kannada (ಕನ್ನಡ), Hindi, English.
**Google AI:** Voice-to-intent parsing extracts the appliance issue, urgency, and address, pre-populating the booking in 1 tap for elderly/rural users.

### 4. Proximity & Pool Isolation Guard

**Feature:** Strictly enforces a 25km geofence.
**Fraud Block:** A technician in Bangalore cannot see, accept, or finish jobs in Sagara.
**Pool Isolation:** Once a job is accepted, public pool jobs are hidden until the job is completed or cancelled.

### 5. Admin Mission Control, Doorstep Audits & SOS Emergency Center

**Customer Privacy vs Admin Audit:** Customers only see verified technician badges; raw site photos and selfies are securely routed to the private `doorstep_audits` collection.
**SOS Siren:** Real-time audio siren in `/admin` with instant 112 emergency dialing and customer/technician telephony routing.

## 5 Step-by-Step Implementation Roadmap

1. **Step 1: Free Gemini API Key Setup** — Acquire free API key from `aistudio.google.com` and add `VITE_GEMINI_API_KEY` to `frontend/.env`.
2. **Step 2: Create Resilient `gemini.ts` Engine** — Build rate-limited client-side wrapper calling `gemini-1.5-flash` with structured JSON outputs and offline fallback.
3. **Step 3: Integrate Visual Diagnostic into Customer Booking** — Add photo upload trigger with real-time AI fault breakdown and transparent parts pricing.

4. **Step 4: Connect Live Selfie to Gemini Biometric Face Match** — Perform multimodal similarity comparison during technician doorstep arrival and record into Firestore `doorstep_audits`.
5. **Step 5: Voice Assistant (Web Speech + Gemini Intent)** — Enable 1-tap Kannada/English voice intake on search bar.
6. **Step 6: Automated Vitest Test Suite Execution** — Verify that all 18+ test suites pass with 100% coverage.
7. **Step 7: Production Build & Firebase Hosting Deploy** — Run `npm run build` and `npx firebase deploy --only hosting:myurbandaily`.

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## 6 Winning 10-Slide Pitch Deck Blueprint

Judges evaluate the pitch deck (PDF) on problem alignment, scalability, and technical elegance:

| SLIDE # | SLIDE TITLE | KEY CONTENT & WINNING VISUAL HOOK |
|---|---|---|
| **Slide 1** | **UrbanAI: The 360° Trusted Gig Economy** | Vision statement: Transforming hyperlocal services in Tier-2/Tier-3 emerging markets with Google Cloud GenAI. |
| **Slide 2** | **The Field Service Trust Deficit** | The 3 critical failures: Impersonation fraud, out-of-region dispatch failures (Bangalore vs Sagara), and opaque pricing. |
| **Slide 3** | **Our Solution: 4 Pillars of AI Trust** | Multimodal Visual Diagnostics + 25km Geofenced Radar + Biometric KYC Face Match + Real-time SOS Dispatch. |
| **Slide 4** | **Google Cloud Architecture** | High-res system diagram highlighting Firebase Hosting, Cloud Firestore, Gemini 1.5 Flash, and Cloud Run. |
| **Slide 5** | **Multimodal Visual Diagnostics** | Screenshot: Customer uploads photo of leaking water purifier → Gemini outputs fault, parts, and transparent ₹350 quote. |
| **Slide 6** | **Doorstep Verification & Biometric KYC** | Mandatory Step 1 Physical Arrival (<500m) + Gemini 1.5 Pro face comparison (Aadhaar KYC vs Live Selfie) + 4-digit OTP. |
| **Slide 7** | **Tier-2 & Tier-3 Market Inclusivity** | Sagara, Hosanagara, Honnali operational hubs; Kannada/Hindi voice booking empowering elderly and rural customers. |
| **Slide 8** | **Business Model & Unit Economics** | 10% platform commission + transparent catalog markups + digital warranty certificate upsells. |
| **Slide 9** | **Live Traction & Production Readiness** | Deployed live at `myurbandaily.web.app`, 18 automated test suites passing, real-time Firestore synchronization. |
| **Slide 10** | **The Team & Scaling Vision** | Engineering pedigree, expansion plan across Karnataka & pan-India semi-urban clusters. |

## 7 Official 3-Minute Video Demo Script

| TIMESTAMP | SCREEN ACTION | SPOKEN NARRATION (SCRIPT) |
|---|---|---|
| **0:00 - 0:35** | Show Sagara Map & Problem Slide | *"In India's emerging towns, booking a home service is plagued by fraud: fake technicians arrive, contractors in distant cities take jobs they cannot fulfill, and pricing is completely opaque. Meet UrbanAI, powered by Google Cloud."* |
| **0:35 - 1:15** | Customer App: Snap photo of faulty appliance | *"A customer in Sagara snaps a photo of their leaking water purifier. Gemini 1.5 Flash instantly inspects the image, diagnoses the faulty membrane, lists replacement parts, and gives a fixed, transparent estimate. No hidden fees."* |
| **1:15 - 1:55** | Technician App: Proximity radar & Step 1 Arrival | *"Our Google Cloud geofence guarantees that only verified technicians within 25km can see the job. Upon arriving within 500m, Step 1 triggers automatically, sending an instant arrival alert to the customer's phone."* |
| **1:55 - 2:35** | Doorstep Selfie & Biometric KYC Face Match | *"Before entering the home, the technician takes a live selfie. Gemini 1.5 Pro performs facial biometric matching against their government KYC ID, verifying liveness and blocking spoofing. The customer shares their 4-digit OTP to start the job."* |
| **2:35 - 3:00** | Admin Mission Control & SOS Emergency Siren | *"In Admin Mission Control, dispatchers monitor real-time audit logs and live SOS alerts with one-click 112 emergency routing. UrbanAI: Built on Google Cloud, empowering the next billion users."* |

### 🎯 HACKATHON SUBMISSION CHECKLIST

- ✅ **Live Web App URL:** `https://myurbandaily.web.app` (Deployed)
- ✅ **GitHub Repository:** `https://github.com/kparunsagar/urbanai` (18 Test Suites Passing)
- ✅ **Target Track:** Retail & Commerce: Intelligent Customer and Business Experiences
- ✅ **Deadline:** Prototype Submission: October 18, 2026 | Grand Finale in Singapore: December 4, 2026